



# MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : ESC501 Software Engineering

UPID : 005505

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

## Group-A (Very Short Answer Type Question)

I. Answer any ten of the following :

[ 1 x 10 = 10 ]

- (I) Level 0 DFD is also called \_\_\_\_\_.
- (II) Functional testing or behavioral testing is also known as \_\_\_\_\_.
- (III) The COCOMO stands for \_\_\_\_\_.
- (IV) Class, interface, feature, constraint etc are the parts of which UML diagram?
- (V) Which SDLC model results in successive versions of the system?
- (VI) \_\_\_\_\_ between two modules is a measure of the degree of interaction.
- (VII) What is minimal test suite?
- (VIII) Delphi cost estimation technique is an example of a \_\_\_\_\_ model.
- (IX) UML structure diagrams show the \_\_\_\_\_ structure of the system.
- (X) If the code for the prototype model is used in the actual project, it is called \_\_\_\_\_.
- (XI) In a DFD, if two bubbles are directly connected by a data flow arrow, then they are \_\_\_\_\_.
- (XII)  $V(G) = E - N + 2$  is a formula to calculate \_\_\_\_\_.

## Group-B (Short Answer Type Question)

Answer any three of the following :

[ 5 x 3 = 15 ]

2. Briefly explain UML Implementation Diagram with suitable example. 48-49 [5]
3. What is Equivalence Class Partitioning? Design the equivalence class test cases for a program that reads two integer pairs (m1, c1) and (m2, c2) defining two straight lines of the form  $y=mx+c$ . The program computes the intersection point of the two straight lines and displays the point of intersection. [5]
4. Differentiate between Object-oriented and function-oriented design approaches. [5]
5. What is UML Activity Diagram? How is the activity partition shown in the Activity Diagram? [5]
6. Write a short note on Installation Testing. [5]

## Group-C (Long Answer Type Question)

Answer any three of the following :

[ 15 x 3 = 45 ]

7. (a) What are the functions of Software Project Management? 48-49 [6]  
(b) What is Software Configuration Management? [5]  
(c) Briefly discuss the 4 P's of Software Project Management. [4]
8. (a) Write a short note on UML. 48-49 [5]  
(b) Briefly explain Weak Aggregation and Strong Aggregation in class diagrams. [6]  
(c) What is binary and n-ary association in class diagram? [4]
9. (a) Define prototype. What are the different types of prototype? 39 [6]  
(b) Briefly explain the prototyping model of software development. [9]
10. (a) What is Data Flow Diagram? What it is required? 30-31 [4]  
(b) Briefly explain the symbols used for constructing DFDs. [3]  
(c) Create a Data Flow Diagram (only level 1) for the following problem: [8]

An online job portal facilitates jobseekers to apply for a job. The jobseekers have to create a profile using name, email, qualification, dob, gender, resume and photo. The system will validate all input data and generate a unique id for each jobseeker. Then the jobseeker can search for a job by title, profile, company, package etc. All new jobs are added by the system admin. After successfully applied for a job, an application list will be stored in the system with jobseeker id, job id, date of application etc. Finally the jobseekers will be able to view the selection list.

11. (a) Briefly discuss the V-Model of software development with suitable diagram.  
(b) Compare V-model with waterfall model. 79  
(c) Briefly explain the advantages and disadvantages of V-model.

[ 5 ]

[ 4 ]

[ 6 ]

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